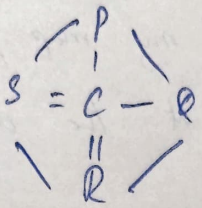
Username: 17sdheeraj

Enigma number: 8411

$(1\times1)+(7\times2)+(19\times4)+(4\times8)+(8\times16)+(5\times32)+(5\times64)$
$+(18\times128)+(1\times256)+(10\times512)=8411$

⇒ Bridges of city



Guess: Initially thought it would be impossible.

Attempts:

1) $C \to P \to Q \to R \to S \to$ ~~P~~
2) $R \to Q \to P \to S \to R \to C \to P$
3) $S \to C \to R \to C \to S \to P \to Q \to R$

⇒ Bridges touching/connecting each other

P	3	Odd
Q	3	Odd
R	4	Even
S	4	even
C	6	even

The walk is possible:

$P \to Q \to R \to S \to P \to C \to R \to C \to S \to C \to Q$

⇒ Why it works:

Every time I walk visits an intermediate island, it consumes bridges in pairs one to enter & one to exit. Coz of this any island not at the start or end must have an even no of bridges.

The starting and finishing islands have one unpaired exit/entrance, this means their counts are odd. Since only 8 & 8 are odd in my map, it must start at one and end at the other.

AI disclosure:

Yes, I used chatgpt to verify my Enigma number arithmatic, confirm the node degrees on map.